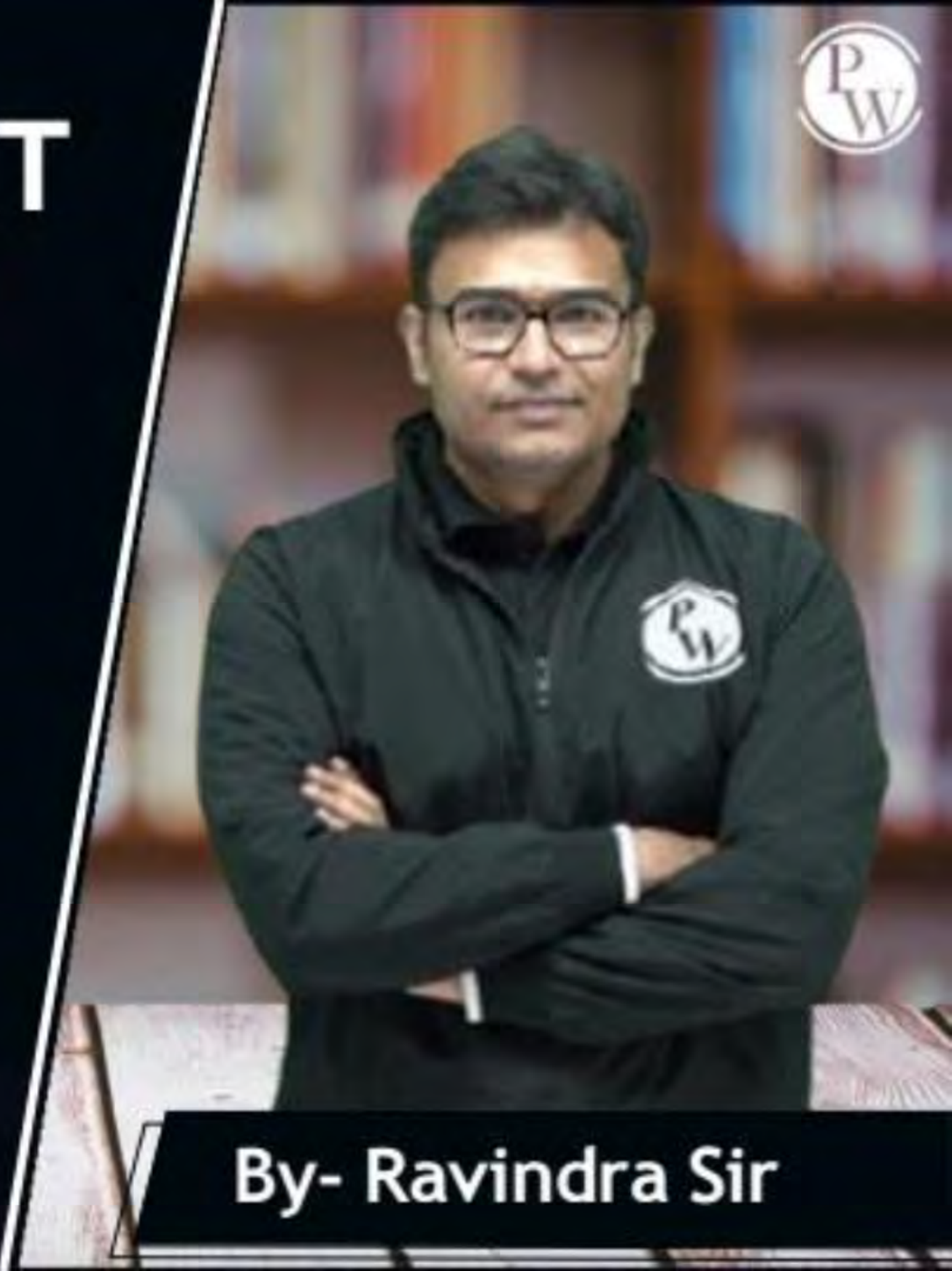
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 09



By- Ravindra Sir

Recap of Previous Lecture



Topic DHCP

Topic

Topic

Topics to be Covered



- Topic — ICMP
- Topic — Traceroute
- Topic — PMTUD

Inspiring Stories : S. Thimmakka



Background: Daily wage worker from Karnataka.

Education: No formal schooling.

Achievements: Planted hundreds of banyan trees along a dry road.

Impact: Turned a barren land green, inspired community tree planting.

Inspiring Stories : Arunima Sinha



Background: Former volleyball player from Uttar Pradesh.

Education: Continued studies after a train accident.

Achievements: Lost a leg, then climbed Mount Everest with a prosthetic.

Impact: Became a face of unstoppable spirit for youth.

Inspiring Stories : William K.



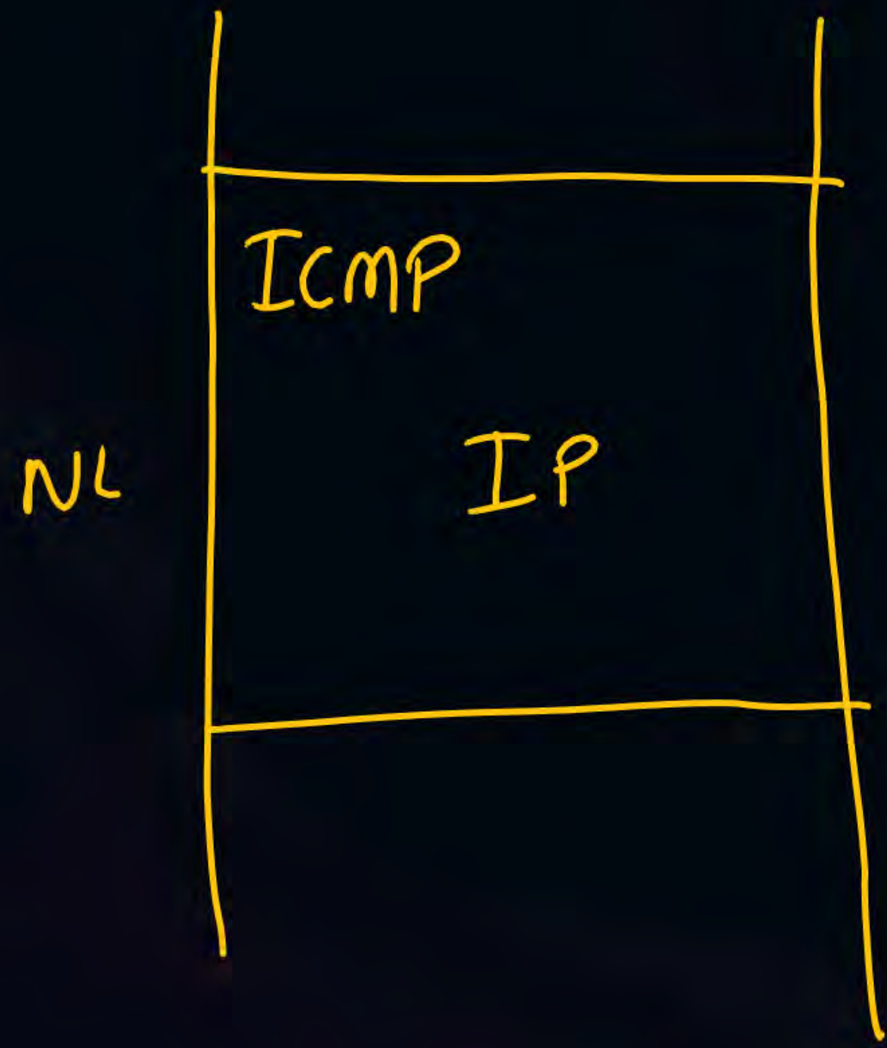
Background: Farmer's son in Malawi, family faced drought and hunger.

Education: Dropped out due to fees, learned from library books.

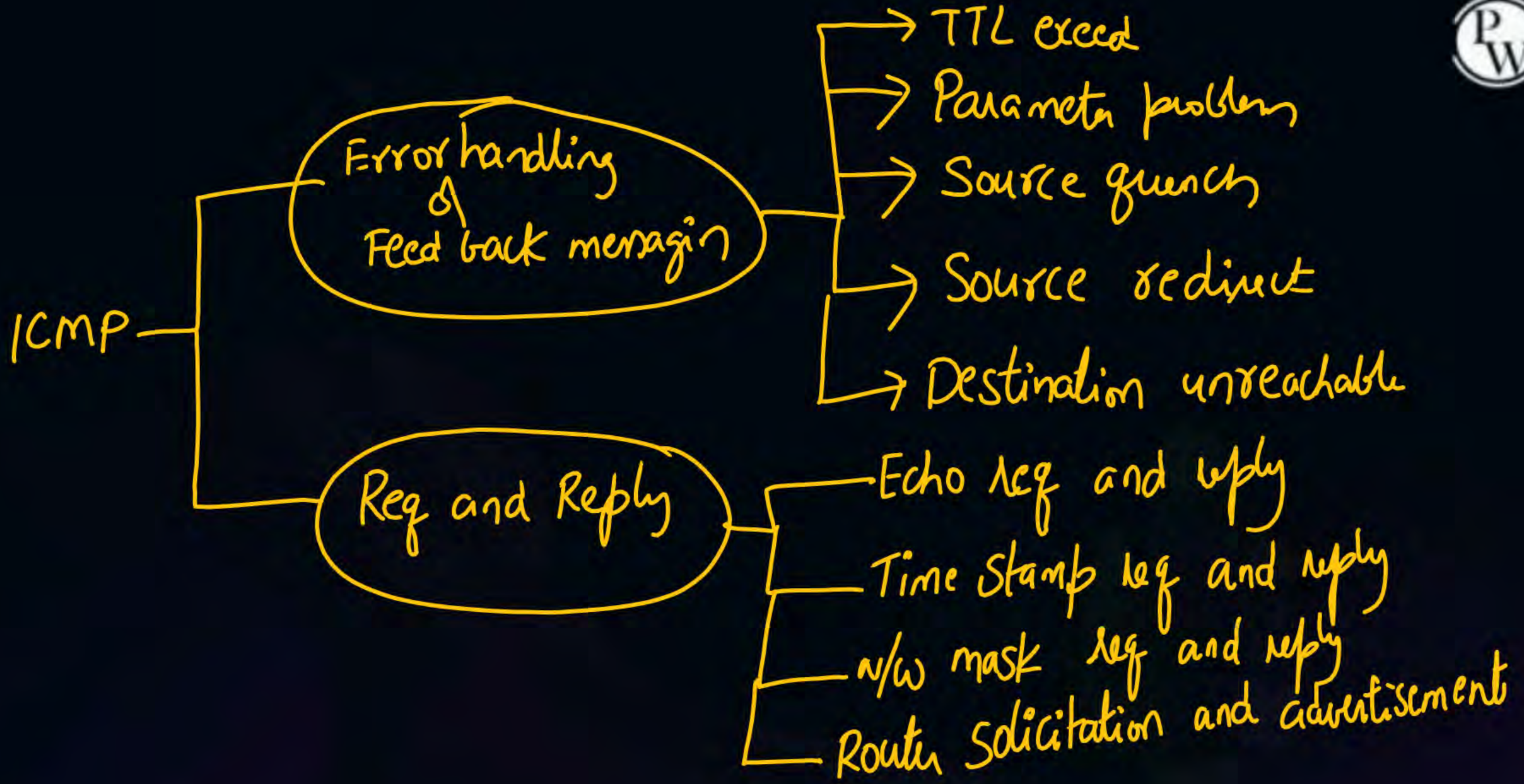
Achievements: Built a windmill from scrap to power his home.

Impact: Brought light to his village, inspired student inventors worldwide.

ICMP



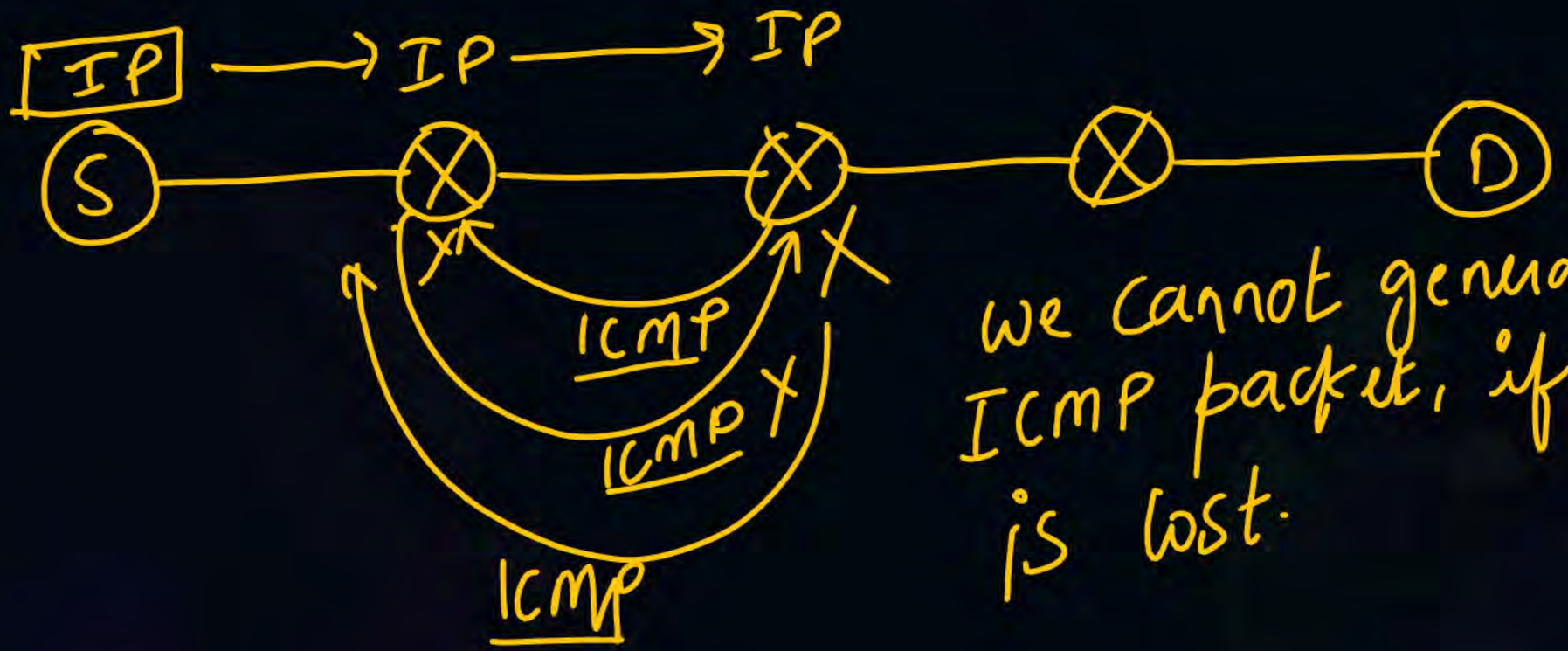
ICMP travels inside IP


ICMP
Error handling
&
Feed back messagin
TTL exced
Parameter problem
Source quench
Source redirect
Destination unreachable
Req and Reply
Echo req and reply
Time Stamp req and reply
n/w mask req and reply
Router Solicitation and advertisement

Error handling :- whenever there is an error, error handling messages will come

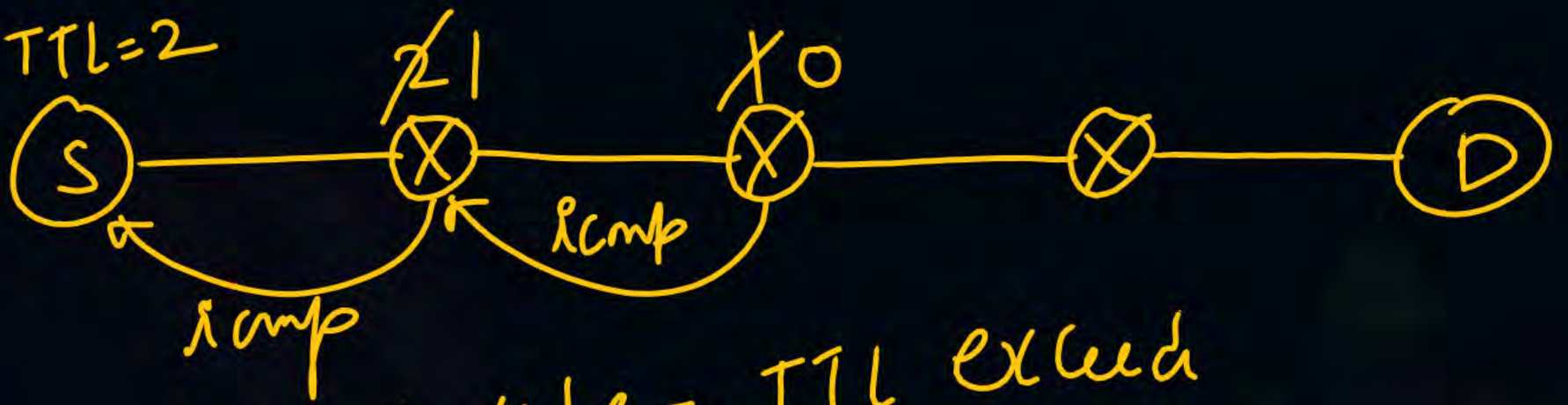
Req and Rep :- we can initiate an ICMP packet

IP + ICMP → Still unreliable



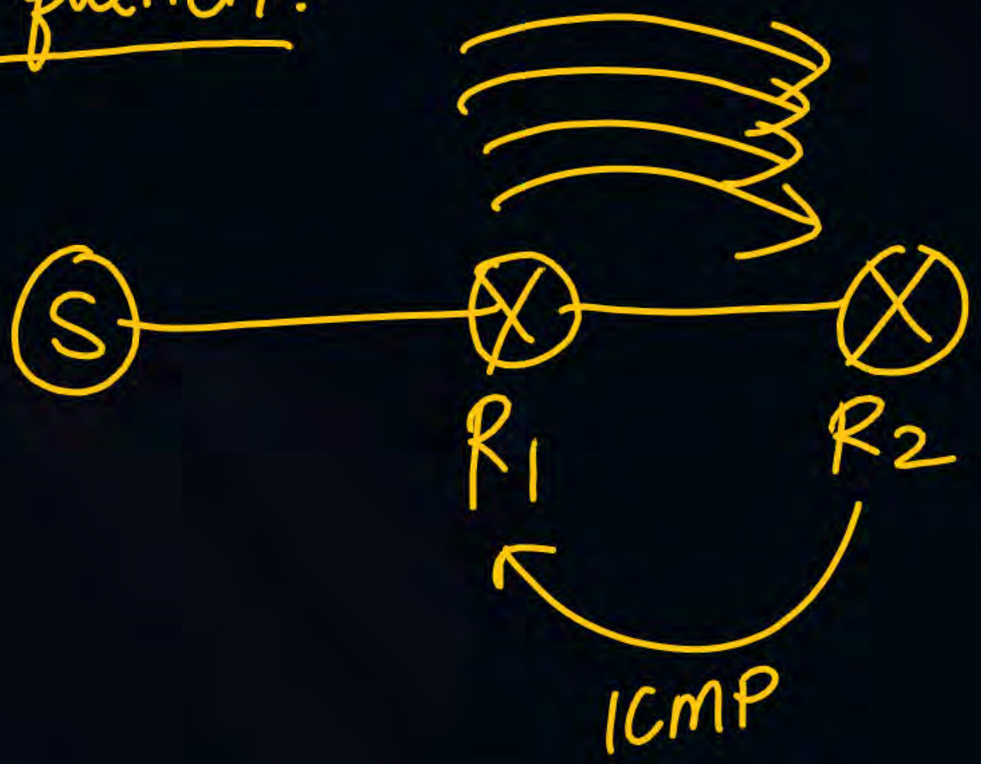
we cannot generate an ICMP packet, if ICMP is lost.

TTL exceed :-



icmp type = TTL exceed

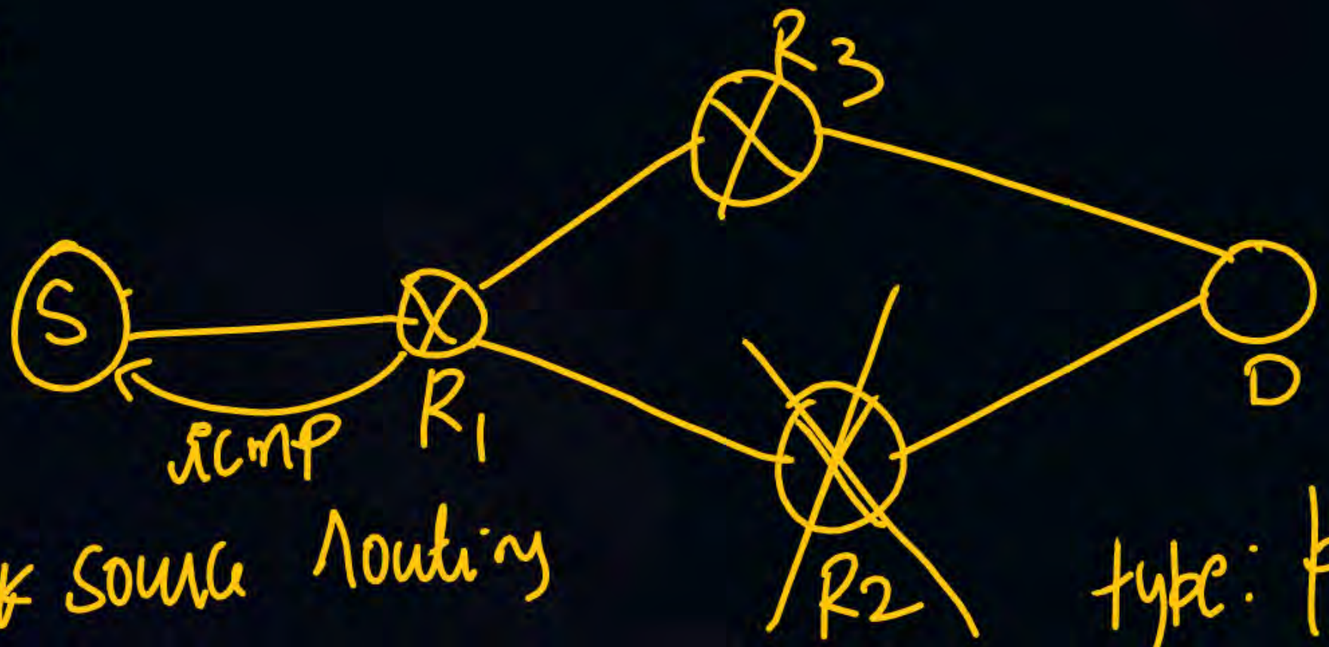
Source quench :-



Type : Source quench

Parameter problem :-

(1)



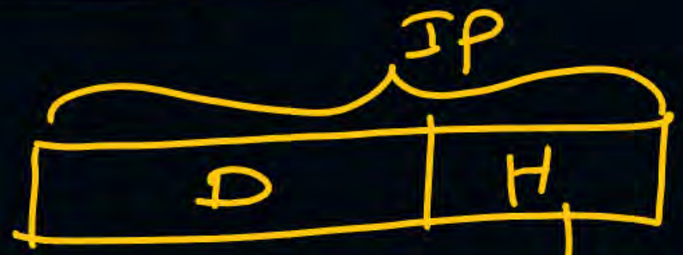
Strict source routing

type: parameter problem

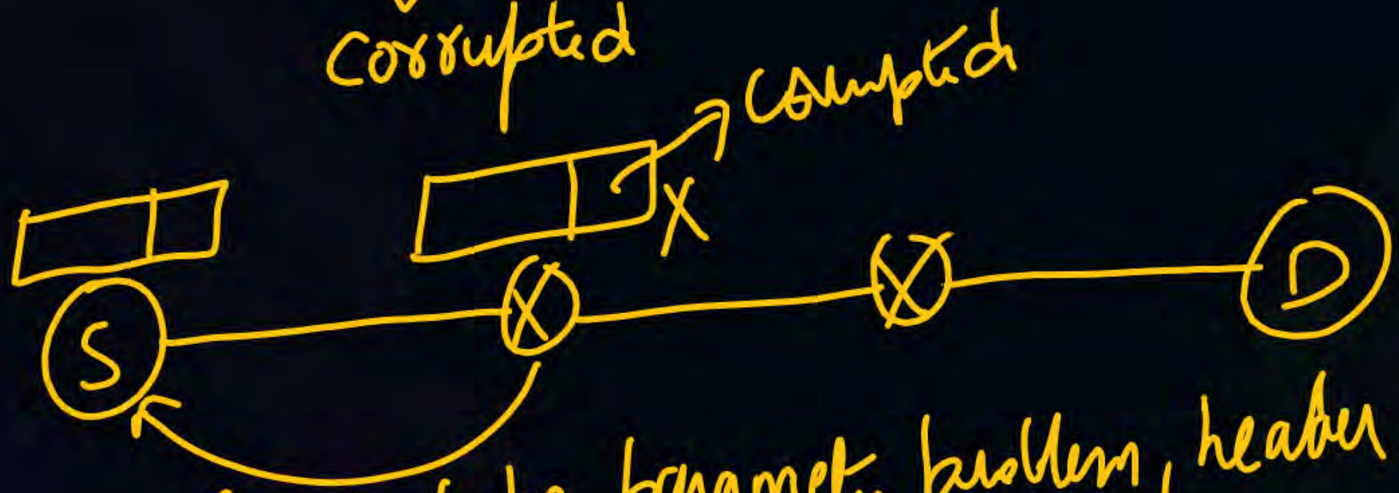


Parameter problem :-

(2)

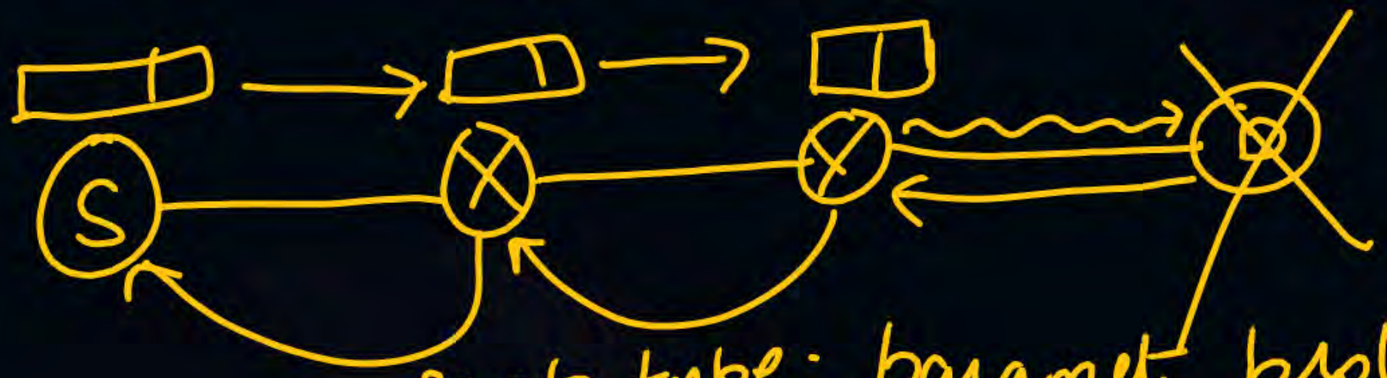


corrupted



icmp type parameter problem, header got corrupted

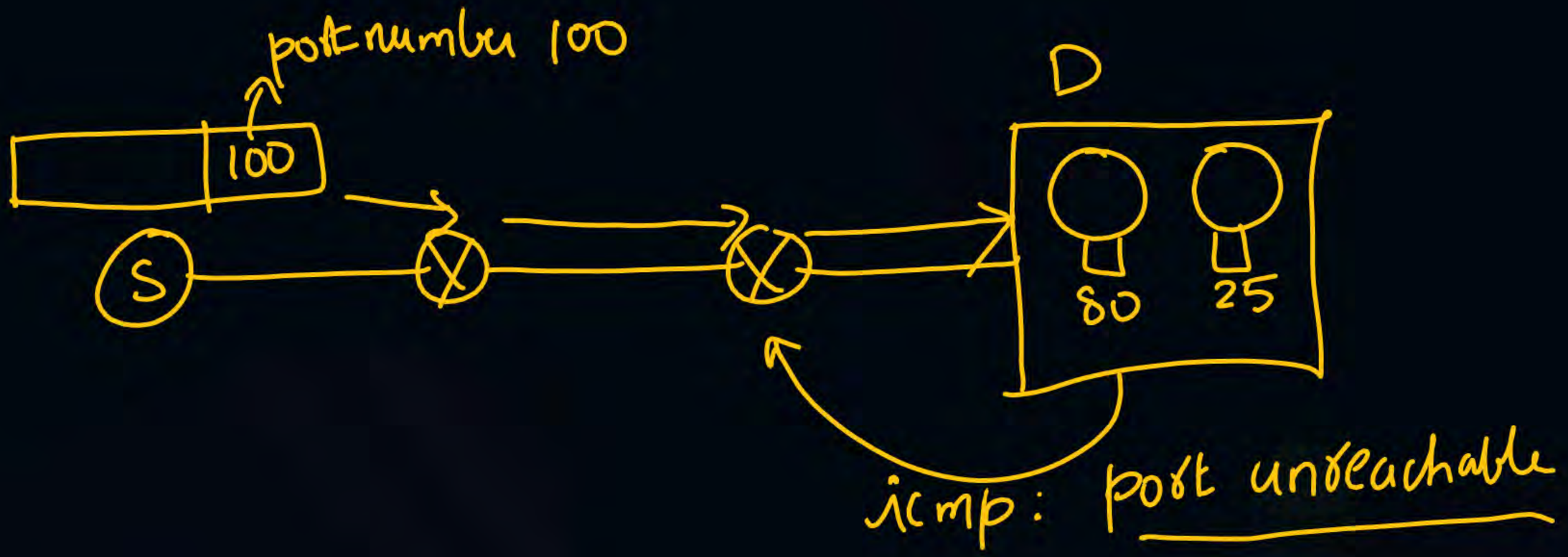
Destination unreachable :-



icmp type: parameter problem

Destination unreachable

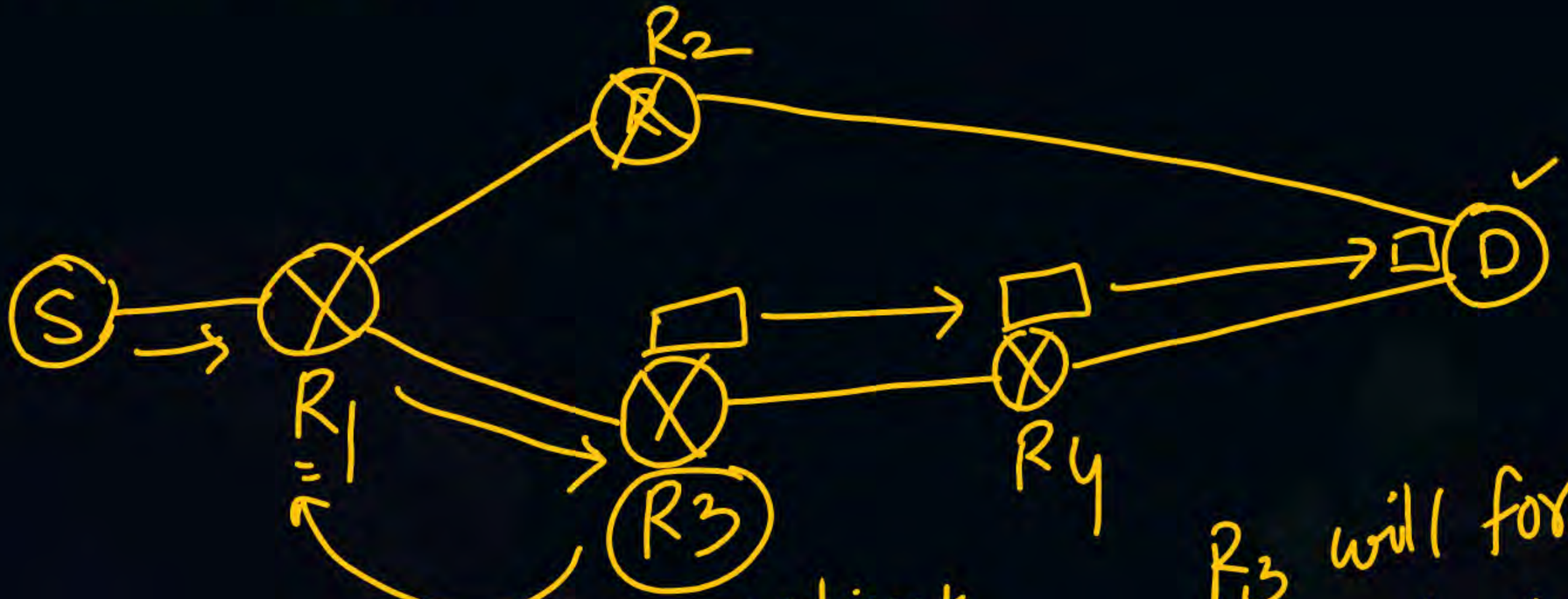
- Destination Host unreachable
- Destination Port unreachable


port number 100
100
S
D
80
25
icmp: port unreachable

one more type of destination unreachable ✓



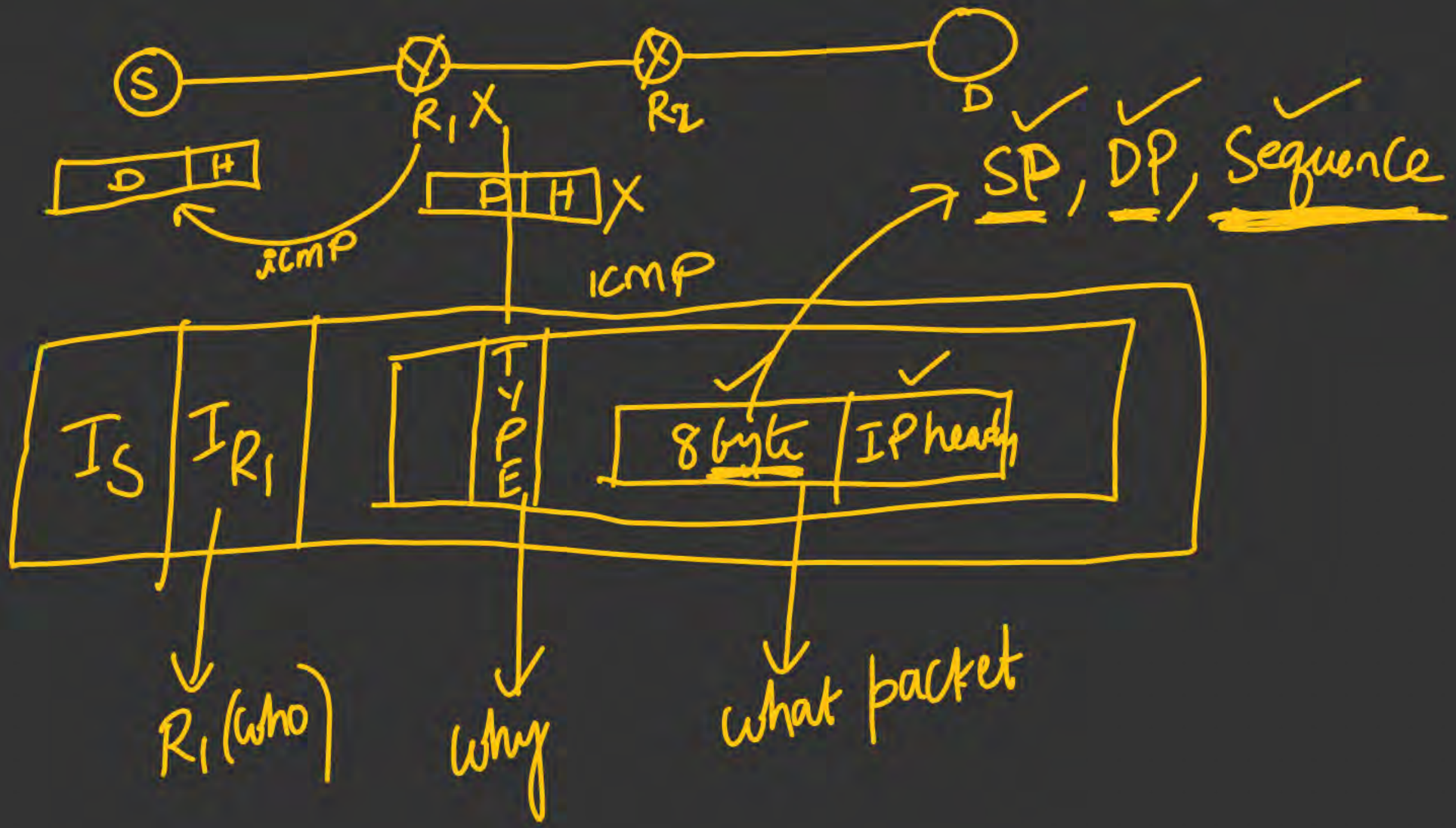
Source Redirect :-



ICMP: Souce redirect warning

R_3 will forward the packet

When even an ICMP packet comes, 1) who 2) why 3) what

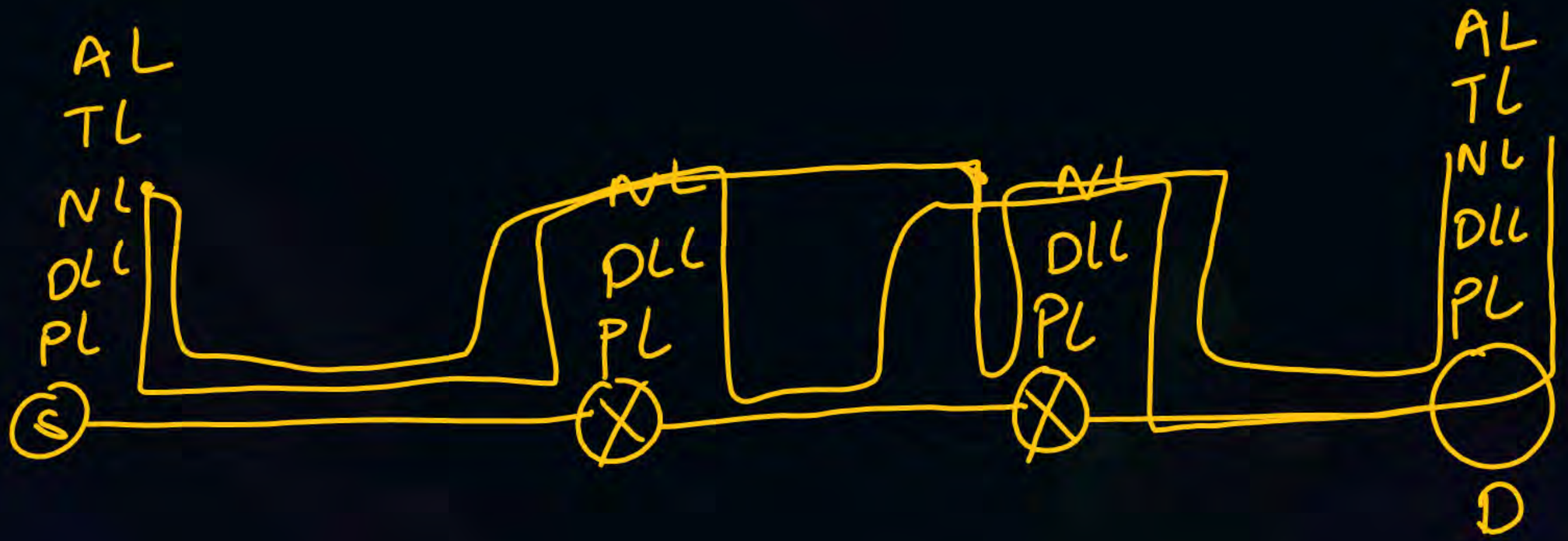


Note:- ICMP is generated only for TCP and UDP.

In case of fragmentation, ICMP will be generated only for the I fragment.

Offset = 0

Echo req and reply



Echo Req and Reply:-



It is used to test if NL of Destination is working or not and also all the NL of the routers on the way are working or not.

Ping command uses echo req and reply

Ping is not client server Application.

DOS attack can be done using echo req and reply

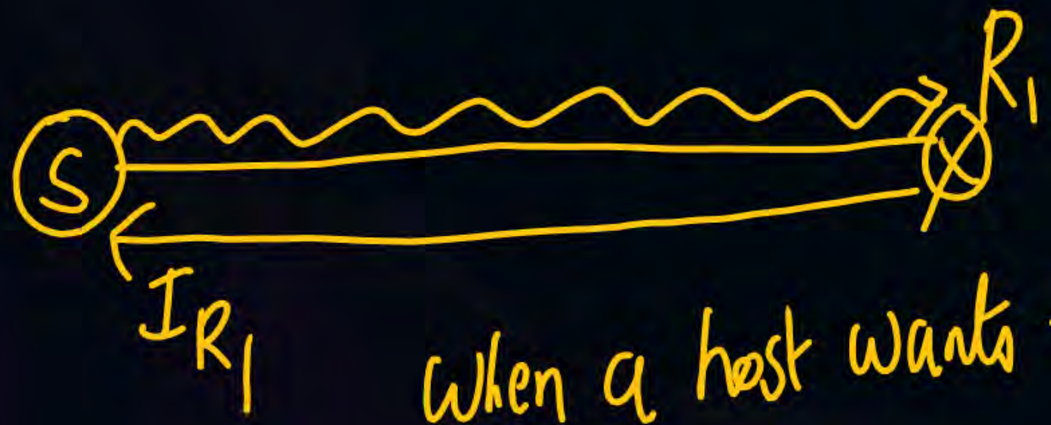


Router Solicitation :-

IP → DHCP

DGW of Router → ICMP

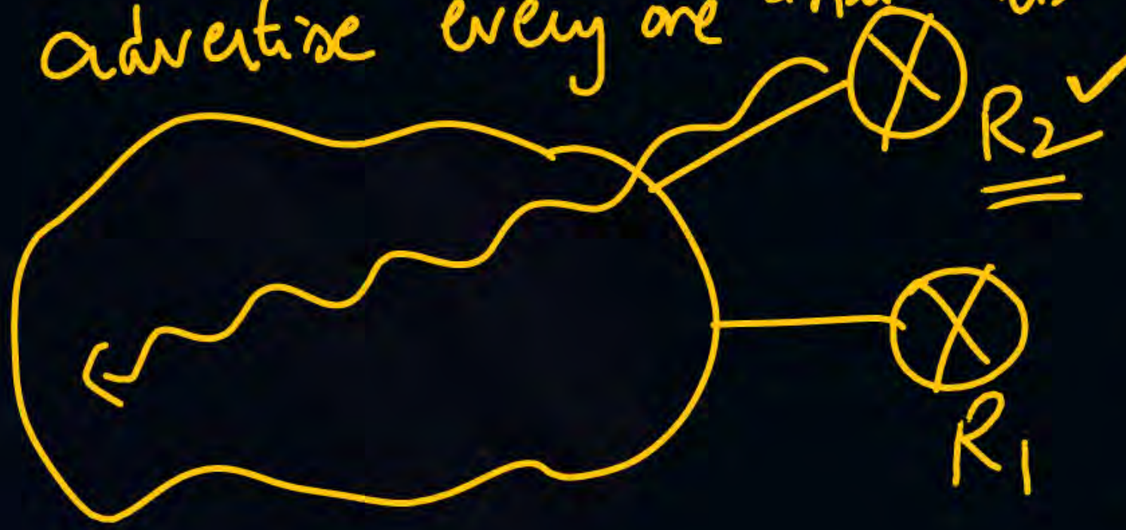
Sm → ICMP



When a host wants to know IP addr of default Router.

Router advertisement

When a new router is connected to a N/w, it will advertise every one that it is availble



IP → DHCP

DR → ICMP

SM → ICMP

N/w mask req and reply :-

When a host wants to know what is the N/w mask of the N/w, it will broadcast an ICMP packet of type N/w mask req and reply



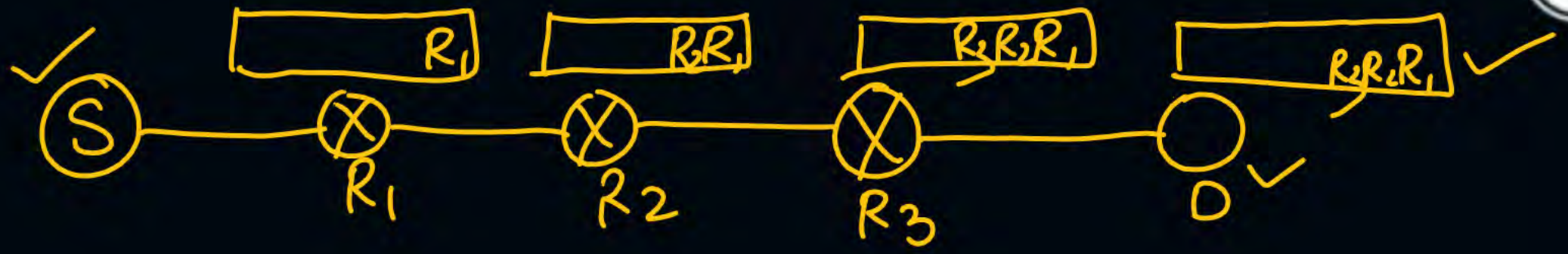
Time Stamp Req and Reply:-

N/w devices are placed at different parts of the world

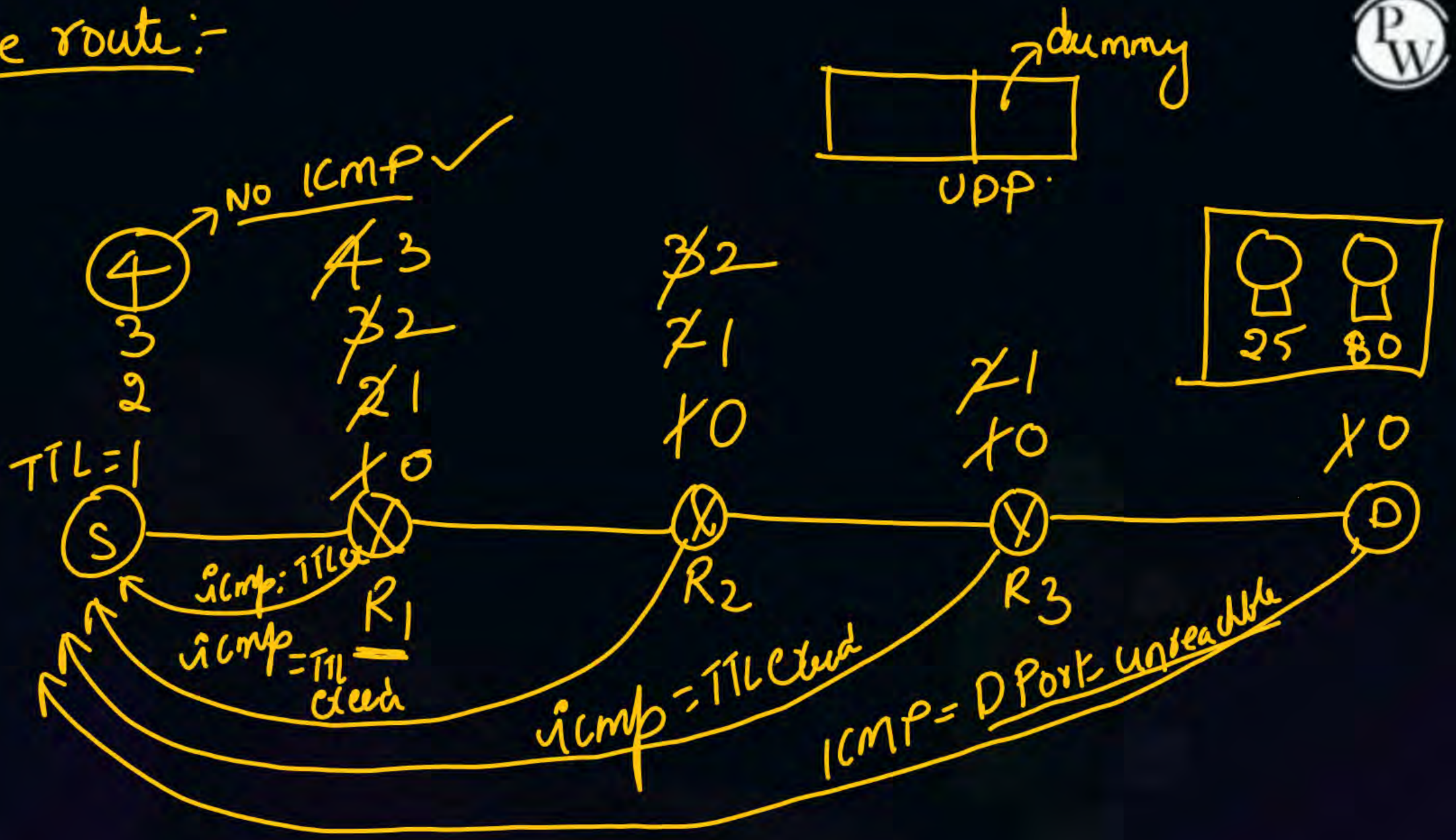
Time zone will be different.

Time stamp Req and Reply are used to

Synchronise the clocks.

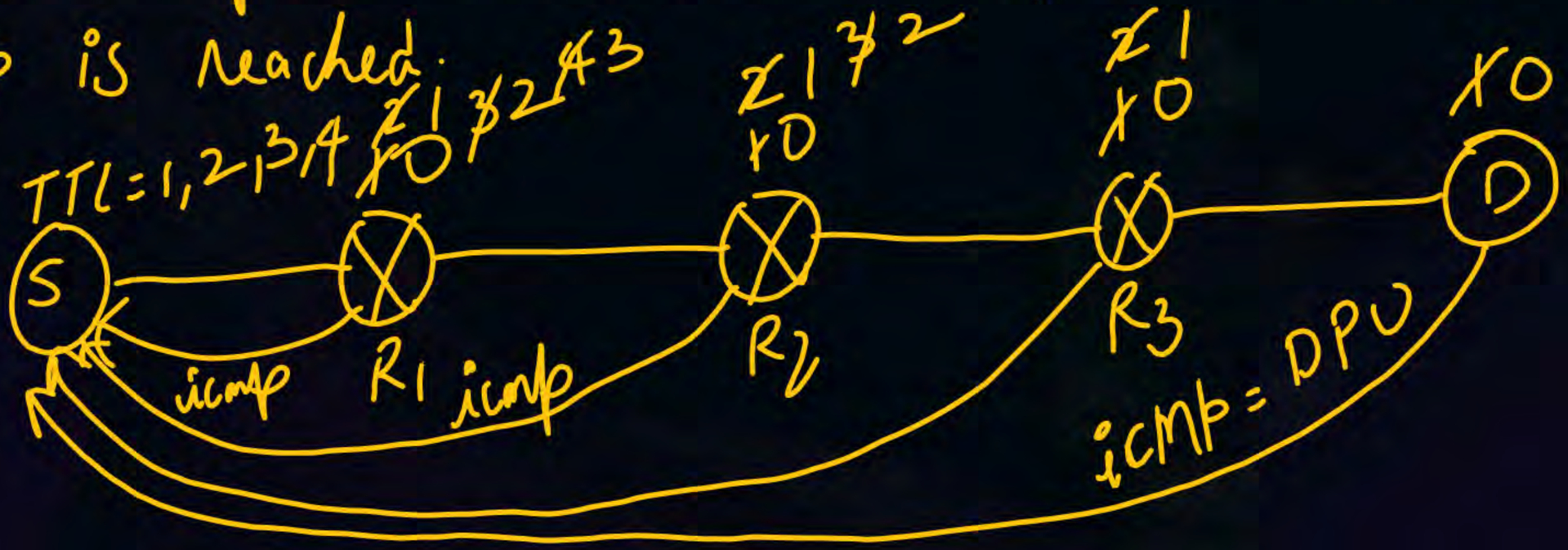

S
R1
R2
R3
D
R1
R2R1
R3R2R1
R4R3R2R1

Trace route :-

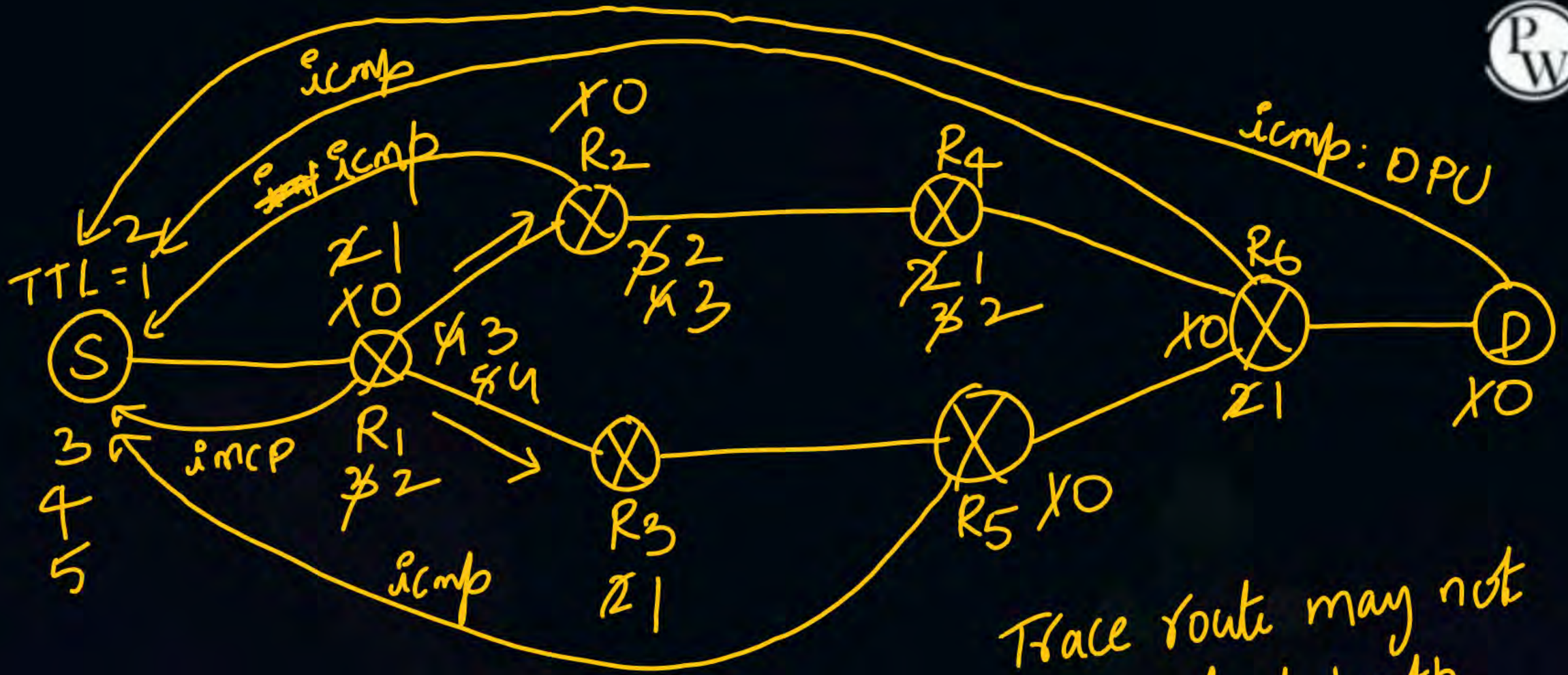

dummy
UDP
NO ICMP ✓
4
3
2
TTL=1
S
A 3
Z 2
Z 1
X 0
R1
icmp: TTL
icmp = TTL Used
Z 2
Z 1
X 0
R2
Z 1
X 0
R3
X 0
D
25
80
icmp = TTL Used
ICMP = D Port Unreachable

Implimentation of trace route:

1) Source will Send a UDP packet with dummy port numbu
2) Initially TTL = 1, and it will increase TTL for every packet
3) Step 2 is repeated until destination port unreachable ICMP is reached.



If we dont get ICMP packet, it doesnot mean destination is reached ✓, icmp might have been lost
That is why we are sending UDP packet with dummy Port number.
When destination recieves UDP with dummy port number, it will send an ICMP of type Destination port unreachable



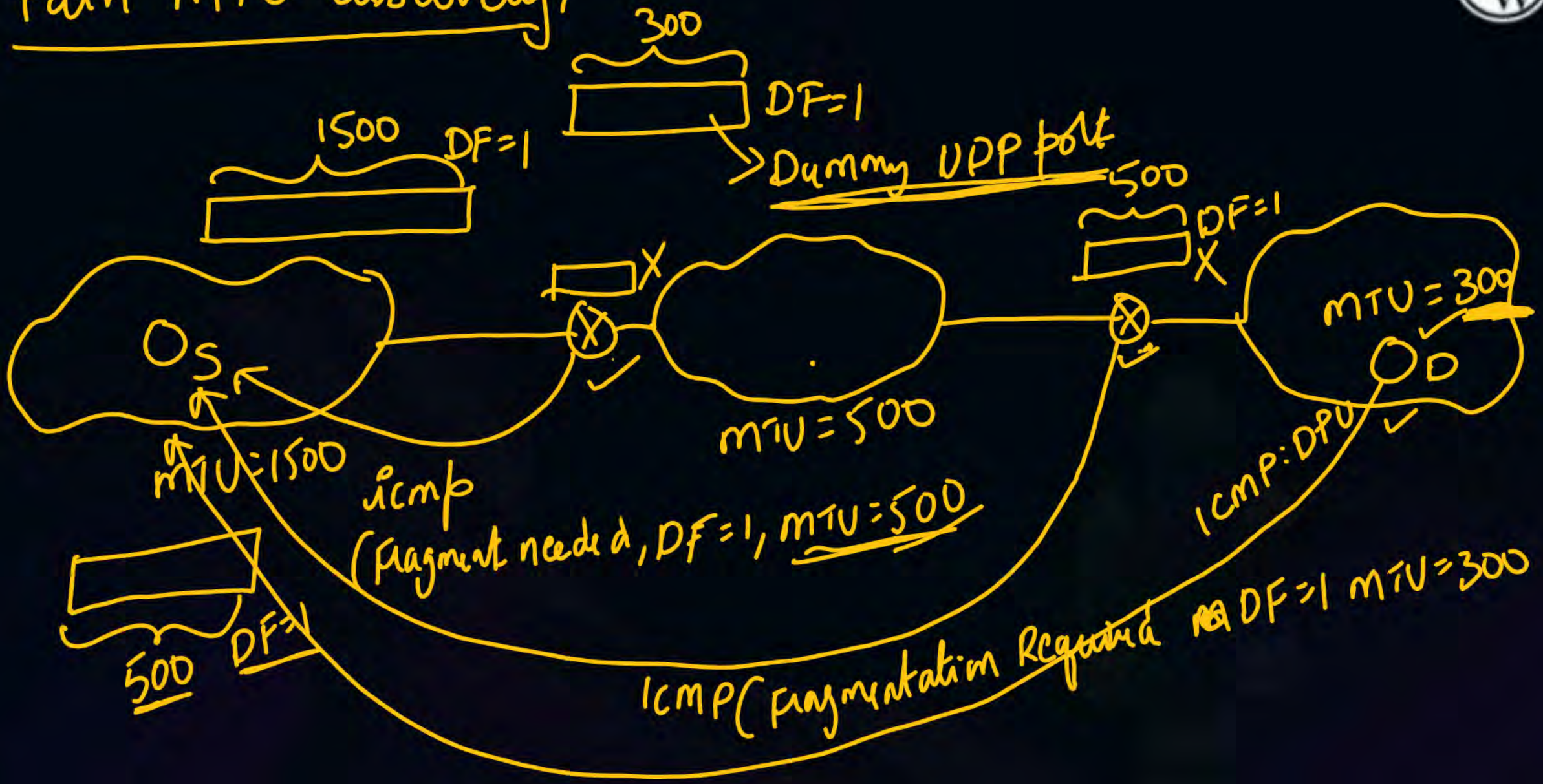
Trace route may not give actual path

$R_1\ R_2\ R_5\ R_6\ D$

Trace route may not give actual path

Because some of the packets may go in different paths.

Path MTU discovery



For above example

1) Sent a dummy UDP packet with dummy port number of size 1500

2) ICMP will with MTU=500, then send 500 Size dummy UDP packet

3) ICMP will come with MTU = 300, then send 300 Size dummy UDP packet

4) ICMP with come with type DPU

So the path MTU = 300

PMTUD → path MTU Discovery

THANK - YOU